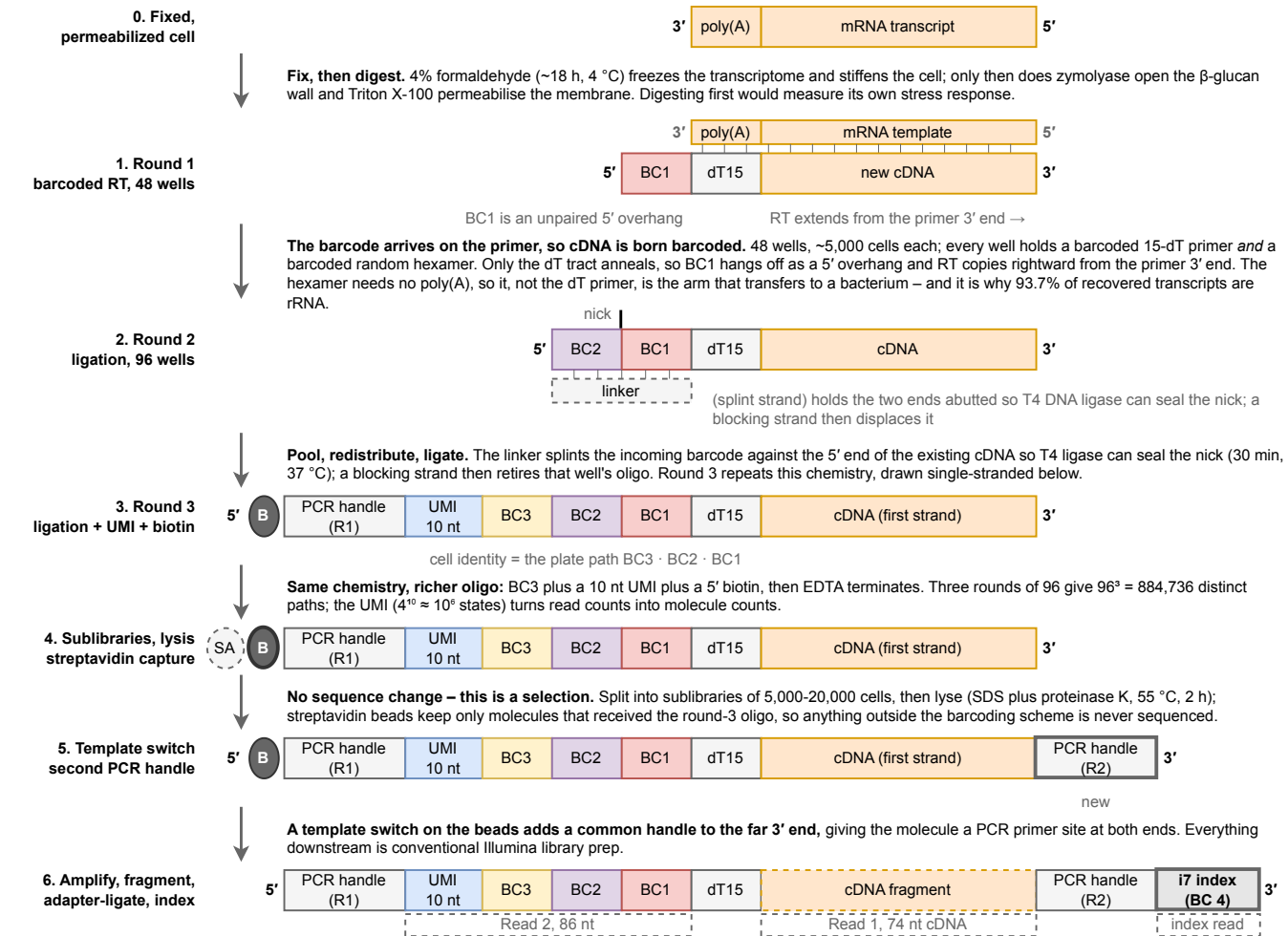


Split-seq (SPLiT-seq): what happens to the sequence at each step

Yeast-optimized SPLiT-seq (Brettner 2024) The tracked strand runs 5' to 3' left to right, so a partner strand runs antiparallel beside it. Steps 1 and 2 are drawn as duplexes because that is where the chemistry acts on two strands.



The Illumina index is barcode round 4 in all but name. The library is amplified from the two handles, fragmented, adapter-ligated and index-PCR'd; the biotin stays behind on the bead. Because cells were split into sublibraries *after* in-cell barcoding, two cells that took identical plate paths are still resolved if they landed in different sublibraries, which is why unique dual indices are mandatory. Three rounds × 24 sublibraries give 21.2 M distinct labels, against 884,736 for the plates alone.

Segments	transcript / cDNA	BC1 (round 1)	BC2 (round 2)	BC3 (round 3)	UMI	handle / adapter	i7 index (round 4)	B	biotin	ll base pairing
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